

UGC NET Economics — PYQs

Unit 10 — Statistical Methods, Mathematics & Econometrics

218 questions · 14 papers (Dec 2018 – Dec 2025) · English only · with options · no answers

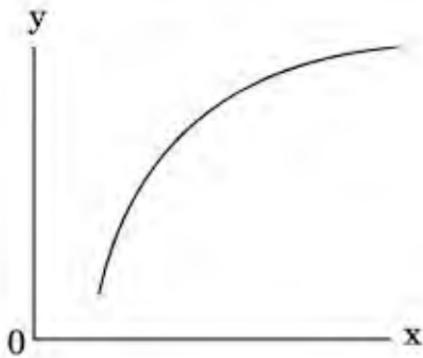
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Source: UGC NET Economics (Paper II) question papers / response sheets in D:\DIVYANSHU\09 UGC NET ECONOMICS\PYQ (Dec 2018 – Dec 2025; duplicate files skipped). Questions are grouped by the UPHESC Assistant Professor Economics syllabus units; original paper and question number shown next to each question. English only · with options · no answers. Micro (Unit 1) questions are in the separate Micro PDF. Dec 2019 paper file ends at Q145 (Q146–150 not in the source).

Unit 10 — Statistical Methods, Mathematics & Econometrics

UGC NET December 2018

Q1. For the diagram given below, which of the following is correct in terms of derivatives? [December 2018 · Q.136]



- (1) $dy/dx > 0$, $d^2y/dx^2 < 0$
- (2) $dy/dx > 0$, $d^2y/dx^2 > 0$
- (3) $dy/dx < 0$, $d^2y/dx^2 < 0$
- (4) $dy/dx < 0$, $d^2y/dx^2 > 0$

Q2. Match List I with List II and choose the correct answer from the code given below: [December 2018 · Q.140]

List - I	List - II
(a) Hawkins-Simon Conditions	(i) Theory of Games
(b) Saddle Point	(ii) Non-Linear Programming
(c) Simplex Method	(iii) Input-Output Analysis
(d) Kuhn-Tucker Conditions	(iv) Linear Programming

- (1) (a)-(i), (b)-(ii), (c)-(iv), (d)-(iii)
- (2) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)
- (3) (a)-(iii), (b)-(ii), (c)-(i), (d)-(iv)
- (4) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)

Q3. A box contains 10 bulbs, 4 of which are defective. If 3 of the bulbs are drawn randomly without replacement, what is the probability that all three bulbs are defective? [December 2018 · Q.142]

- (1) $1/30$
- (2) $3/4$
- (3) $3/10$
- (4) $4/10$

Q4. A series of index numbers is given as follows: [December 2018 · Q.143]

Year	2014	2015	2016
Index Number	100	120	126

If the base year is shifted to year 2015, the value of index number in the year 2016, in the new series will be:

- (1) 106
- (2) 95.2
- (3) 105
- (4) 151.2

Q5. In a two-variable regression, the correlation coefficient between dependent and independent variables Y and X is 0.7. What is the percentage of variations in Y explained by X? [December 2018 · Q.144]

- (1) 49%
- (2) 1%
- (3) 0.7%
- (4) 70%

Q6. Consider the following statements about a time series of annual observations on sales of a firm for ten years: [December 2018 · Q.145]

- (i) The series will have seasonal variations.
- (ii) The series will have a secular trend.
- (iii) The series will have cyclical variations.

Choose the correct answer from the code given below:

- (1) (ii) and (iii)
- (2) (i), (ii) and (iii)
- (3) (i) and (ii)
- (4) (i) and (iii)

Q7. Match List I with List II and choose the correct answer from the code given below: [December 2018 · Q.146]

List - I	List - II
(a) Delphi Method	(i) Stationarity
(b) Dickey-Fuller Test	(ii) Autocorrelation
(c) Durbin-Watson Test	(iii) Testing significance of a regression coefficient
(d) Student's t-Test	(iv) Forecasting

- (1) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)
- (2) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)
- (3) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)
- (4) (a)-(i), (b)-(iv), (c)-(iii), (d)-(ii)

Q8. By autocorrelation we mean that [December 2018 · Q.147]

- (1) The residuals of a regression model are not independent.
- (2) The residuals of a regression model are related with one or more of the regressors.
- (3) The square of the residuals of a regression model are not equally spread.
- (4) The variance of the residuals of a regression model is not constant for all observations.

Q9. In Regression Analysis, if a quantitative variable has 'm' categories, one can introduce [December 2018 · Q.148]

- (1) Only $m + 1$ dummy variables
- (2) Only $m - 1$ dummy variables
- (3) Only m dummy variables
- (4) Only 2 m variables

Q10. Though two time series are individually non-stationary, their linear combination is stationary. This is an example of [December 2018 · Q.149]

- (1) Random walk
- (2) Spurious regression
- (3) Cointegration
- (4) Trend stationarity

Q11. In the context of simultaneous equations econometric models, consider the following statements: [December 2018 · Q.150]

- (i) Both order and rank conditions for identification are to be satisfied before estimation.
- (ii) Order condition is a necessary but not a sufficient condition of identification.
- (iii) Two-stage least squares method is suitable for over-identified equations.

Which of the above statements is/are correct?

Choose the correct answer from the code given below:

- (1) Only (i)
- (2) (i), (ii) and (iii)
- (3) (i) and (iii)
- (4) Only (ii)

UGC NET June 2019 (21 June 2019, Shift 1)

Q12. In a binomial distribution, the probability of x successes in n trials in any order is given by [June 2019 S1 · Q.134]

- (1) $p(x) = nCx p^x \cdot q^{n-x}$
- (2) $p(x) = nC1 p^{2x} \cdot q^{n-1}$
- (3) $p(x) = nCx p^2 \cdot q^{n-1}$
- (4) $p(x) = nCx p^x \cdot q^{n-2}$

Q13. The correct relationship among Arithmetic Mean (AM), Geometric Mean (GM) and Harmonic Mean (HM) is [June 2019 S1 · Q.135]

- (1) $GM = AM + HM$
- (2) $AM < HM$
- (3) $[GM]^2 = HM \times AM$
- (4) $AM = GM - HM$

Q14. For a moderately skewed distribution, the value of mode is 10 and value of median is 20. Then the mean is [June 2019 S1 · Q.136]

- (1) 15
- (2) 30
- (3) 25
- (4) 35

Q15. Standard Error (SE) of sample mean ($\bar{x}$) in random sampling from an infinite population or in random sampling with replacement is given by [June 2019 S1 · Q.137]

- (1) $SE = \sigma^2/\sqrt{n}$ where σ is the population standard deviation
- (2) $SE = \sigma/\sqrt{(n^2)}$ where σ is the population standard deviation
- (3) $SE = \sigma^3/\sqrt{n}$ where σ is the population standard deviation
- (4) $SE = \sigma/\sqrt{n}$ where σ is the population standard deviation

Q16. When the null hypothesis is there as an equality, the level of significance is the probability of making [June 2019 S1 · Q.138]

- (1) type II error
- (2) type I error
- (3) type I and II errors
- (4) type I and III errors

Q17. In a two variable (X and Y) regression analysis, the regression coefficients are $b_{xy} = 0.6$ and $b_{yx} = 0.8$. If y is dependent variable and x is independent variable, what is the percentage of variations in Y explained by X? [June 2019 S1 · Q.139]

- (1) 47%
- (2) 48%
- (3) 49%
- (4) 36%

Q18. Match List-I with List-II : [June 2019 S1 · Q.140]

List - I	List - II
(a) Gauss-Markov	(i) R^2
(b) Coefficient of determination	(ii) One sided test
(c) $H_0 : B_2 = 0, H_1 : B_2 > 0$ [printed as " $= 70$ "]	(iii) Goldfeld-Quandt test
(d) Heteroscedasticity	(iv) BLUE

Choose the correct option from those given below :

- (1) (a)-(iv); (b)-(i); (c)-(ii); (d)-(iii)
- (2) (a)-(i); (b)-(ii); (c)-(iv); (d)-(iii)
- (3) (a)-(ii); (b)-(iii); (c)-(i); (d)-(iv)
- (4) (a)-(iii); (b)-(i); (c)-(ii); (d)-(iv)

Q19. Multiplicative decomposition model is used to [June 2019 S1 · Q.141]

- (1) deseasonalise the time series data
- (2) build cost of living index
- (3) test a hypothesis
- (4) estimate probability

Q20. Match List-I with List-II : [June 2019 S1 · Q.142]

List - I	List - II
(a) Skewness	(i) Dispersion
(b) Variance	(ii) Random variation
(c) Earthquake	(iii) Ratio of variances
(d) F-test	(iv) Lack of symmetry

Choose the correct option from those given below :

- (1) (a)-(iii); (b)-(iv); (c)-(i); (d)-(ii)
- (2) (a)-(iv); (b)-(i); (c)-(ii); (d)-(iii)
- (3) (a)-(i); (b)-(iii); (c)-(iv); (d)-(ii)
- (4) (a)-(ii); (b)-(i); (c)-(iv); (d)-(iii)

Q21. If $A = \begin{bmatrix} 6 & 5 & 5 \\ 3 & -11 & -5 \\ 3 & -5 & -2 \end{bmatrix}$ (a 3×3 matrix, rows separated by ;), then $|A|$ is equal to

[June 2019 S1 · Q.143]

- (1) 15
- (2) 16
- (3) 27
- (4) 20

Q22. Difference equation is concerned with [June 2019 S1 · Q.145]

- (1) discrete changes
- (2) continuous changes
- (3) Both of the above
- (4) None of the above

Q23. A linear programming problem can be solved by using [June 2019 S1 · Q.146]

- (1) ARIMA techniques
- (2) Simplex method
- (3) Unit root method
- (4) None of the above

Q24. In an open economy input-output model, the viability is examined by [June 2019 S1 · Q.147]

- (1) Hawkins-Simon conditions
- (2) Kuhn-Tucker conditions
- (3) Dickey-Fuller test
- (4) Goldfeld-Quandt test

Q25. Applying OLS to simultaneous equations results in the parameters being [June 2019 S1 · Q.148]

- (1) inefficient
- (2) inconsistent
- (3) biased
- (4) biased and inconsistent

Q26. A simultaneous equation model is exactly identified, if [June 2019 S1 · Q.150]

- (1) unique numerical values of structural parameters can be obtained
- (2) more than one numerical value can be obtained for some of the parameters of the structural equation
- (3) unique solution of all the structural coefficients is not possible
- (4) structural coefficient cannot be estimated

UGC NET December 2019 (3 Dec 2019, Shift 1)

Q27. For an exactly identified equation, the order condition that should be fulfilled is that equation must: [December 2019 S1 · Q.59]

- (1) Exclude one less than the total number of endogenous variables in the model
- (2) Include one less than the total number of endogenous variables in the model
- (3) Include only one of the endogenous variables in the model
- (4) Exclude two or more endogenous variables

Q28. Analysis of variance is carried out by using: [December 2019 S1 · Q.60]

- (1) t-test
- (2) Z-test
- (3) χ^2 test
- (4) F-test

Q29. Under Normal Probability Curve, the area within the range of $\mu - 1.96\sigma$ and $\mu + 1.96\sigma$ (where μ = mean, σ = standard deviation) is: [December 2019 S1 · Q.61]

- (1) 90%
- (2) 95%
- (3) 75%
- (4) 99%

Q30. Boot strapping technique is used to: [December 2019 S1 · Q.62]

- (1) test for specification bias
- (2) obtain sampling distribution of parameters of interest
- (3) test for auto-correlation
- (4) test for normality of error term

Q31. Inverse of matrix $\begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ is (matrices written row by row, rows separated by ";") [December 2019 S1 · Q.63]

- (1) $\begin{bmatrix} 0.25 & 0 & 0 & 0 \\ 0 & 0.5 & 0 & 0 \\ 0 & 0 & 0.2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
- (2) $\begin{bmatrix} 0.2 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & .25 & 0 \\ 0 & 0 & 0 & 0.5 \end{bmatrix}$
- (3) $\begin{bmatrix} 0.5 & 0 & 0 & 0 \\ 0 & 0.25 & 0 & 0 \\ 0 & 0 & 0.2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
- (4) $\begin{bmatrix} 0.5 & 0 & 0 & 0 \\ 0 & 0.25 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0.4 \end{bmatrix}$

Q32. Three statement are given as under: [December 2019 S1 · Q.95]

- (a) Generalised Least Square (GLS) method is capable of providing BLUE in situations where OLS fails
- (b) GLS is OLS on the transformed variables that satisfy the standard least squares assumptions
- (c) GLS method is suitable for dealing with multicollinearity problem

Choose the correct option:

- (1) (a) and (b) are correct but (c) is not correct
- (2) (a) and (c) are correct but (b) is not correct
- (3) (a) is correct but (b) and (c) are not correct
- (4) (a) is not correct but (b) and (c) are correct

Q33. Which of the following statements are true concerning a triangular or recursive system?

[December 2019 S1 · Q.96]

- (a) The parameters can be validly estimated using separate applications of OLS to each equation
- (b) The independent variables may be correlated with the error terms in other equation
- (c) An application of two-stage least squares would lead to unbiased but inefficient parameter estimation
- (d) The independent variables may be correlated with the error terms in the equation in which they appear as independent variables

Choose the correct option:

- (1) (a) and (d) only
- (2) (a) and (c) only
- (3) (a), (b) and (c) only
- (4) (a), (b), (c) and (d)

Q34. Two series are given as under [December 2019 S1 · Q.97]

- (a) $1 - 1/2 + 1/4 - 1/8 + 1/16 - 1/32 + \dots$ upto ∞
(b) $2 + \sqrt{2} + 1 + 1/\sqrt{2} + 1/2 + 1/(2\sqrt{2}) + \dots$ upto ∞

Choose the correct option:

- (1) (a) is convergent while (b) is divergent series
(2) Both (a) and (b) are convergent series
(3) (a) is divergent while (b) is convergent series
(4) None of (a) and (b) are convergent

Q35. In Linear Programming each problem (called primal) has a dual. About duality in Linear Programming following statements are given: [December 2019 S1 · Q.98]

- (a) If a Linear Programming problem is maximizing type, then its dual is minimizing type
(b) Both primal and dual have same optimum value
(c) Dual of dual of a linear programming problem is primal problem
(d) If maximizing linear programming problem is with "less than" type constraints, the dual of this problem will have "more than" type constraints

Choose from the following options:

- (1) Only (a) and (b) are correct
(2) Only (c) and (d) are correct
(3) (a), (b) and (c) are correct
(4) All (a), (b), (c), (d) are correct

Q36. Match the tests with their purpose given below: [December 2019 S1 · Q.113]

List - I	List - II
(a) Dickey-Fuller Test	(i) Heteroscedasticity
(b) Box-Jenkins Method	(ii) Causality
(c) Glejser Test	(iii) Forecasting
(d) Granger Test	(iv) Stationarity

Choose the correct option from those given below:

- (1) (a)-(iii), (b)-(i), (c)-(ii), (d)-(iv)
(2) (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)
(3) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)
(4) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)

Q37. Match the Economists with their field of contribution in Mathematical Economics: [December 2019 S1 · Q.114]

List - I	List - II
(a) Slutsky	(i) Linear Programming
(b) Arrow	(ii) Input-output analysis
(c) Leontief	(iii) Consumer behaviour analysis
(d) Dantzig	(iv) CES Production Function

Choose the correct option from those given below:

- (1) (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)
(2) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)
(3) (a)-(i), (b)-(ii), (c)-(iv), (d)-(iii)
(4) (a)-(iv), (b)-(i), (c)-(iii), (d)-(ii)

Q38. In terms of their contribution to Econometrics which is correct chronological order: [December 2019 S1 · Q.123]

- (1) L.R. Klein, H. Theil, A.L. Nagar, T. Haavelmo
- (2) A.L. Nagar, T. Haavelmo, L.R. Klein, H. Theil
- (3) T. Haavelmo, L.R. Klein, H. Theil, A.L. Nagar
- (4) H. Theil, L.R. Klein, A.L. Nagar, T. Haavelmo

Q39. Which is correct chronological order of Economists contributing to input-output analysis?

[December 2019 S1 · Q.124]

- (1) Hawkins-Simon, Leontief, Karl Marx, Quesnay
- (2) Karl Marx, Quesnay, Leontief, Hawkins-Simon
- (3) Quesnay, Karl Marx, Leontief, Hawkins-Simon
- (4) Leontief, Hawkins-Simon, Karl Marx, Quesnay

Q40. Given below are two statements — one is labelled as Assertion (A) and the other is labelled as Reason (R): [December 2019 S1 · Q.133]

Assertion (A): Geometric Mean is theoretically considered as the best average to Construct Index Numbers

Reason (R): Geometric mean is used if the range of observations are extremely large

In the light of the above two statements, choose the correct option:

- (1) Both (A) and (R) are true and (R) is correct explanation of (A)
- (2) Both (A) and (R) are true and (R) is not correct explanation of (A)
- (3) (A) is true but (R) is false
- (4) (A) is false but (R) is true

UGC NET June 2020 (exam held 30 Sep 2020, Shift 2)

Q41. For any two equally likely, exhaustive and independent events A and B, $P(AB)$ will be equal to

[June 2020 S2 · Q.75]

- (1) 0
- (2) 0.25
- (3) 0.50
- (4) 1

Q42. What type of relationship exists between two variables, if all points on the scatter diagram appear to fall on a straight line going upward from left to right? [June 2020 S2 · Q.76]

- (1) Positive correlation
- (2) Perfect positive correlation
- (3) Negative correlation
- (4) Perfect negative correlation

Q43. Given ' σ ' as the Standard Deviation of a given series and if to each and every observation of the series is added a constant ' k ', then what will be the Standard Deviation of the new series? [June 2020 S2 · Q.77]

[June 2020 S2 · Q.77]

- (1) $\sigma + k$
- (2) $\sigma - k$
- (3) σ
- (4) σk

Q44. Which one of the following price Index numbers satisfy the Factor Reversal Test? [June 2020 S2 · Q.78]

- (1) Laspeyres Price Index Number
- (2) Paasche's Price Index Number
- (3) Walsh's Price Index Number
- (4) Fisher's Price Index Number

Q45. Match List I with List II [June 2020 S2 · Q.79]

List - I	List - II
List I (Statistic)	List II (Standard Error)
(A) Sample Mean ($\bar{x}$)	(I) $\sqrt{[(\sigma_1^2/2n_1) + (\sigma_2^2/2n_2)]}$
(B) Sample Variance (s^2)	(II) $\sqrt{[(\sigma_1^2/n_1) + (\sigma_2^2/n_2)]}$
(C) Difference of two independent sample means ($\bar{x}_1 - \bar{x}_2$)	(III) $\sigma/\sqrt{n}$
(D) Difference of two independent sample standard Deviations ($s_1 - s_2$)	(IV) $\sigma^2\sqrt{(2/n)}$

Choose the correct answer from the options given below:

- (1) A - III, B - IV, C - II, D - I
- (2) A - III, B - II, C - IV, D - I
- (3) A - III, B - IV, C - I, D - II
- (4) A - I, B - III, C - II, D - IV

Q46. Given the sample size 400 with sample mean 99, the population mean 100 and computed value of z - statistic at - 2.5, the value of population standard deviation will be [June 2020 S2 · Q.80]

- (1) 59.17
- (2) 123.46
- (3) 8
- (4) Cannot be determined

Q47. If the disturbance term is heteroscedastic, which one of the responses based on given statements is true? [June 2020 S2 · Q.81]

- (A) OLS estimates are biased
- (B) OLS estimates do not have the minimum variance property
- (C) Tests of significance based on OLS estimates will be inaccurate
- (D) OLS estimates are inconsistent

Choose the correct answer from the options given below:

- (1) A and B are true only
- (2) A and C are true only
- (3) B and C are true only
- (4) A and D are true only

Q48. Based on the order condition, state, whether the first equation is identified? [June 2020 S2 · Q.82]

Given the following structural model.

$$y_1 = 3y_2 - 2x_1 + x_2 + u_1$$

$$y_2 = y_3 + x_3 + u_2$$

$$y_3 = y_1 - y_2 - 2x_3 + u_3$$

where y 's are the endogenous variables, x 's are predetermined variables and u 's are random terms

- (1) Over Identified
- (2) Under Identified
- (3) Exactly Identified
- (4) Information not sufficient

Q49. Which one of the following response is true as a solution for simultaneous equation bias? [June 2020 S2 · Q.83]

- (A) OLS Method
- (B) Principle Component Method
- (C) Two-Stage Least Square Method (2 SLS Method)
- (D) Full Information Maximum Likelihood Method (FIML)

Choose the correct answer from the options given below:

- (1) A and B are true only
- (2) B and C are true only
- (3) C and D are true only
- (4) A and D are true only

Q50. A time series, under the condition $E(x_t) = \mu$ and $\text{Cov}(x_t, x_{t+k}) = \gamma(k)$ is said to be [June 2020 S2 · Q.84]

- (1) Strictly Stationary
- (2) Non Stationary
- (3) Weakly Stationary
- (4) Can't be defined

Q51. Match List I with List II [June 2020 S2 · Q.85]

List - I	List - II
(A) Linear Function	(I) $y = 3x^2 + 2x + 9$
(B) Quadratic Function	(II) $y = 5x^3$
(C) Exponential Function	(III) $y = 3e^{2x}$
(D) Power Function	(IV) $y = 5x + 3$

Choose the correct answer from the options given below:

- (1) A - IV, B - I, C - III, D - II
- (2) A - IV, B - I, C - II, D - III
- (3) A - IV, B - II, C - III, D - I
- (4) A - I, B - II, C - III, D - IV

Q52. Given below are two statements, one is labelled as Assertion A and the other is labelled as Reason R [June 2020 S2 · Q.88]

Assertion A: If A is any Matrix given by $A = \begin{bmatrix} 5 & 0 & 3 \\ -1 & 0 & 2 \\ 1 & 0 & 1 \end{bmatrix}$ (rows) then $\det A = 0$, since all elements in column II are zero

Reason R: Laplace expansion permits evaluation of a determinant along any row or column

In light of the above statements, choose the most appropriate answer from the options given below

- (1) Both A and R are correct and R is the correct explanation of A
- (2) Both A and R are correct but R is NOT the correct explanation of A
- (3) A is correct but R is not correct
- (4) A is not correct but R is correct

Q53. Which one of the following is a valid objective function as a linear programming problem?

[June 2020 S2 · Q.90]

- (1) $\text{Max } Z = 5x^{2y}$
- (2) $\text{Max } Z = 3x + 2y + 8xy$
- (3) $\text{Min } Z = 2x^2 + 3y^2$
- (4) $\text{Min } Z = 5x + 2y$

Q54. If no industry (sector) draws its own output as input, then the principal diagonal element in the Technological Coefficient Matrix will be [June 2020 S2 · Q.91]

- (1) -1
- (2) +1
- (3) Zero
- (4) Infinity

Q55. Which of the following are Hawkins-Simon conditions? [June 2020 S2 · Q.92]

- (A) Determinant A should be positive
- (B) Determinant (I-A) should be positive
- (C) Determinant (I-A) should be negative
- (D) Principal diagonal elements of Matrix (I-A) should be positive

Choose the correct answer from the options given below:

- (1) A and C are true only
- (2) A and D are true only
- (3) B and D are true only
- (4) C and D are true only

UGC NET December 2020 & June 2021 (merged cycle, exam Nov 2021)

Q56. Assuming the probability of a male birth as 0.5, find the chances that a family of 3 children will have two boys and one girl. [December 2020 · Q.11]

- (1) $1/8$
- (2) $2/8$
- (3) $3/8$
- (4) $4/8$

Q57. Which one of the following measures of central tendency will be the most appropriate to use if the data relates to rates, proportions and ratios? [December 2020 · Q.12]

- (1) Arithmetic Mean
- (2) Median
- (3) Harmonic Mean
- (4) Geometric Mean

Q58. The sampling distribution of mean based on a sample of 13 units selected without replacement shows a variance of 5 units. If the population consists of 49 units, estimate the variance of the population. [December 2020 · Q.13]

- (1) 80.67
- (2) 84.87
- (3) 86.67
- (4) 86.87

Q59. In the classical linear regression model, X_i and U_i are assumed to be: [December 2020 · Q.14]

- (1) highly correlated
- (2) not correlated
- (3) positively correlated
- (4) negatively correlated

Q60. Identify which one of the following statements is INCORRECT? [December 2020 · Q.15]

- (1) Intersection of two disjoint sets is a null set
- (2) Complement of a Universal set is a null set
- (3) If $X \cap Y = \emptyset$, then $X - Y \neq X$
- (4) $X - Y = X \cap \bar{Y}$

Q61. Given $Y = X^3 - 12X^2 + 36X + 8$, identify which one of the following is CORRECT: [December 2020 · Q.16]

- (1) Y is maximum at $X = 6$ and minimum at $X = 2$
- (2) Y is minimum at $X = 6$ and maximum at $X = 2$
- (3) Y has two inflexion points at $X = 2$ and $X = 6$
- (4) Point of optimality is indeterminate

Q62. Identify which one of the following statements is CORRECT: [December 2020 · Q.17]

- (1) A^{-1} is equal to $\text{Adj}(A)$ if the value of the determinant A is unity
- (2) If two columns of a determinant are interchanged, the sign of the determinant does not alter
- (3) The value of a determinant changes if the rows and columns are interchanged
- (4) Law of division is allowed in matrix operations

Q63. Identify which of the following statements with respect to Linear Programming Problem (LPP) is CORRECT: [December 2020 · Q.18]

- (1) A solution is unbounded in an LPP if all the feasible solutions of the problem fall within the boundary created by its constraints
- (2) If A is a primal problem and B is its dual, then dual of B is not A
- (3) The constraints of an LPP in standard form is always expressed in less than and equal to form
- (4) If an LPP has two optimum solutions, then it has also infinite number of optimum solutions

Q64. Which of the following are CORRECT? [December 2020 · Q.45]

- (A) If observations are selected purposively, it is called a method of non-random sampling
- (B) If observations are selected in an ad-hoc manner, it is called a method of non-random sampling
- (C) If observations are selected in a systematic manner it is called a method of non-random sampling
- (D) If observations are selected on the basis of judgment, it is called a method of non-random sampling

Choose the correct answer from the options given below:

- (1) A, B and C only
- (2) B, C and D only
- (3) A, C and D only
- (4) A, B and D only

Q65. Which are the following correct features of matrices? [December 2020 · Q.46]

- (A) An identity matrix is a square matrix
- (B) An identity matrix when gets transposed remains an identity matrix
- (C) $(A.B)^T = A^T.B^T$ where A and B are matrices
- (D) $(A^{-1})^T = (A^T)^{-1}$ where A is a matrix

Choose the correct answer from the options given below:

- (1) A and B only
- (2) B and C only
- (3) A and C only
- (4) C and D only

Q66. Which of the following statements are CORRECT for correlation analysis? [December 2020 · Q.58]

- (A) Degree of linear association is measured between two variables
- (B) Predictability is not part of correlation analysis
- (C) Spearman has no contribution to correlation analysis
- (D) Cause and effect relationship is established through correlation analysis

Choose the correct answer from the options given below:

- (1) A and C only
- (2) B and C only
- (3) A and B only
- (4) A and D only

Q67. Match List I with List II [December 2020 · Q.68]

List - I	List - II
(A) Carl Friedrich Gauss	(I) Central Tendency
(B) Multicollinearity	(II) OLS
(C) Mode	(III) Karl Pearson
(D) Correlation coefficient	(IV) Relationship among explanatory variables

Choose the correct answer from the options given below:

- (1) A - IV, B - I, C - II, D - III
- (2) A - III, B - IV, C - I, D - II
- (3) A - II, B - IV, C - III, D - I
- (4) A - II, B - IV, C - I, D - III

Q68. Match List I with List II [December 2020 · Q.69]

List - I	List - II
(A) Convex Function	(I) Basic Variable
(B) Input-Output Analysis	(II) Use of Derivative
(C) Linear Programming	(III) Hawkins-Simon condition
(D) Marginal Propensity to Consume	(IV) Demand Curve

Choose the correct answer from the options given below:

- (1) A - IV, B - III, C - I, D - II
- (2) A - II, B - III, C - I, D - IV
- (3) A - III, B - IV, C - I, D - II
- (4) A - IV, B - I, C - III, D - II

Q69. Identify the proper sequence in the traditional econometric methodology of conducting research [December 2020 · Q.78]

- (A) Data collection
- (B) Policy prescription
- (C) Testing of hypothesis
- (D) Formulation of hypothesis

Choose the correct answer from the options given below

- (1) D, A, B, C
- (2) A, D, B, C
- (3) D, A, C, B
- (4) A, B, C, D

Q70. Given below are two statements: [December 2020 · Q.90]

Statement I: One of the assumptions under OLS method states that the regression model is linear in the parameters, though it may or may not be linear in the variables.

Statement II: The variance of the error, or disturbance, term is not the same regardless of the value of the explanatory variable under OLS.

In light of the above statements, choose the most appropriate answer from the options given below

- (1) Both Statement I and Statement II are correct
- (2) Both Statement I and Statement II are incorrect
- (3) Statement I is correct but Statement II is incorrect
- (4) Statement I is incorrect but Statement II is correct

UGC NET December 2021 & June 2022 (merged cycle, exam Oct 2022)

Q71. Hari's score in the statistics in the first year is 76, whereas the class average is 68 with a standard deviation of 5. What is Hari's Z-score? [December 2021 · Q.11]

- (1) 1.5
- (2) 1.6
- (3) 1.2
- (4) 1.4

Q72. A bag contains 20 tickets marked with numbers 1 to 20. One ticket is drawn at random. Find the probability that will be multiple of 2 or 5. [December 2021 · Q.12]

- (1) 0.7
- (2) 0.9
- (3) 0.6
- (4) 0.3

Q73. A simultaneous equation model is said to be over-identified if, [December 2021 · Q.13]

- (1) Unique numerical values of the structural parameters can be obtained
- (2) More than one numerical values can be obtained for some of the parameters of the structural equations
- (3) Unique solution of all the structural coefficient is not possible
- (4) Structural coefficient cannot be estimated

Q74. The fitted regression equation is given by $\hat{Y}_i = -10 + 0.6X$. What is the value of the residual at the point $X = 40, Y = 60$. [December 2021 · Q.14]

- (1) 74
- (2) 46
- (3) 0
- (4) 45

Q75. Find the rank of $A = [6 \ 3 \ 5; -10 \ 2 \ 8; 5 \ 2 \ 3]$ (3×3 matrix; rows separated by semicolons) [December 2021 · Q.17]

- (1) 3
- (2) 2
- (3) 1
- (4) 0

Q76. Integrate: $x^2/(x + 2)$ [December 2021 · Q.18]

- (1) $4x/(x + 2)^2 + c$
- (2) $x^2/2 + 2x + 4 \log(x + 2) + c$
- (3) $x^2/2 - 2x + 4 \log(x + 2) + c$
- (4) $x^2/2 - 2x + c$

Q77. Which of the following properties of matrix are correct? [December 2021 · Q.50]

- (A) $(A')' = A$
- (B) The rank of an $n \times n$ nonsingular matrix A must be less than n
- (C) $(AB)' = B'A'$
- (D) $(A + B)^n = A^n + B^n$
- (E) $(A')^{-1} = (A^{-1})'$

Choose the correct answer from the options given below :

- (1) (A), (B) and (C) only
- (2) (A), (D) and (E) only
- (3) (A), (C) and (D) only
- (4) (C), (D) and (E) only

Q78. The binomial distribution can be used under the following conditions [December 2021 · Q.53]

- (A) n , the number of trials is finite and independent
- (B) n , the number of trials is infinite and independent
- (C) Each trial results in two mutually exclusive and exhaustive outcomes, termed as success and failure
- (D) P , the probability of success is constant for each trial
- (E) The probability of failure does not remain same for each trial

Choose the correct answer from the options given below :

- (1) (A), (C) and (E) only
- (2) (B), (C) and (D) only
- (3) (A), (C) and (D) only
- (4) (B), (C) and (E) only

Q79. Which of the followings are correct in case of normal distribution? [December 2021 · Q.54]

- (A) $Q1 + Q2 = 2Md$
- (B) $Q.D. = (1/3)\sigma$
- (C) $Q.D. = (2/3)\sigma$
- (D) All moments of odd order about mean are zero
- (E) $Q1 + Q2 = Md$

Choose the correct answer from the options given below :

- (1) (B), (D) and (E) only
- (2) (A), (B) and (D) only
- (3) (A), (C) and (D) only
- (4) (C), (D) and (E) only

Q80. What are the practical consequences of multicollinearity in CLRM? [December 2021 · Q.56]

- (A) R^2 , the overall measure of goodness of fit can be very high
- (B) OLS estimators are not BLUE
- (C) OLS estimators have small variance and covariances
- (D) t ratio of one or more coefficients is statistically significant
- (E) The OLS estimators and their standard errors can be sensitive to small changes in the data

Choose the correct answer from the options given below :

- (1) (A), (B) and (C) only
- (2) (A), (B) and (D) only
- (3) (A), (D) and (E) only
- (4) (B), (C) and (E) only

Q81. If we have objective function involving n variables : $U = f(X1, X2, X3, \dots, Xn)$ subject to $\phi(X1, X2, X3, \dots, Xn) = 0$. The second order condition for extremum to be [December 2021 · Q.58]

- (A) For maximum $|H2| < 0, |H3| < 0, \dots$
- (B) For maximum $|H2| > 0, |H3| < 0, \dots$
- (C) For maximum $|H2| < 0, |H3| > 0, \dots$
- (D) For minimum $|H2| < 0, |H3| < 0, \dots$
- (E) For minimum $|H2| > 0, |H3| > 0, \dots$

Choose the correct answer from the options given below :

- (1) (A) and (D) only
- (2) (B) and (D) only
- (3) (C) and (E) only
- (4) (B) and (E) only

Q82. Match List I with List II : [December 2021 · Q.69]

List - I	List - II
(A) Testing of overall significance	(I) Chi-square test
(B) Testing the independence of attributes	(II) Jarque-Bera test
(C) Testing the significance of regression coefficients	(III) F-test
(D) Normality test	(IV) t-test

Choose the correct answer from the options given below :

- (1) (A)-(I), (B)-(II), (C)-(III), (D)-(IV)
- (2) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)
- (3) (A)-(I), (B)-(IV), (C)-(III), (D)-(II)
- (4) (A)-(I), (B)-(III), (C)-(IV), (D)-(II)

Q83. Write the steps to estimate the ECM/VAR model in time series. [December 2021 · Q.80]

- (A) Unit root test
- (B) ACF and PACF test
- (C) ECM/VAR
- (D) Inputting data
- (E) Cointegration test

Choose the correct answer from the options given below :

- (1) (D), (A), (E), (B), (C)
- (2) (D), (B), (A), (C), (E)
- (3) (D), (B), (A), (E), (C)
- (4) (B), (A), (D), (E), (C)

UGC NET December 2022 (Shift 1)

Q84. Find the variance of random variables which constitute the number of heads in a toss of two coins simultaneously. [December 2022 S1 · Q.13]

- (1) $1/2$
- (2) 1
- (3) 2
- (4) 4

Q85. Test statistics used to test stationary of a time series in the presence of correlated error terms [December 2022 S1 · Q.16]

- (1) Dickey - Fuller test
- (2) Engle - Granger test
- (3) Error - Correction mechanism
- (4) Augmented Dickey-Fuller test

Q86. In logit model as P_i goes from 0 to 1, logit L varies from [December 2022 S1 · Q.17]

- (1) 0 to $+\infty$
- (2) $-\infty$ to $+\infty$
- (3) 0 to 1
- (4) $-\infty$ to 0

Q87. The regression model $Y_t = \alpha_1 + \alpha_2 D_t + \beta_1 X_t + \beta_2 (Y_t D_t) + U_t$ is used to [December 2022 S1 · Q.19]

- (1) Deseasonalize a time series data
- (2) Analyze the seasonal trend in data
- (3) Analyse structural break in data
- (4) Analyse piecewise linear relationship in data

Q88. A box of nine golf gloves contains two left-handed and seven right-handed gloves. If two gloves are randomly selected from the box without replacement, what is the probability that one is left-handed and the other is a right-handed glove? [December 2022 S1 · Q.20]

- (1) $7/12$
- (2) $2/9$
- (3) $7/18$
- (4) $1/9$

Q89. The properties of Mathematical Expectation are [December 2022 S1 · Q.47]

- (A) $E(C) = 0$, where C is constant
- (B) If X and Y are random variables, then $E(X+Y) = E(X) + E(Y)$
- (C) $\text{Var}(X \pm C) = \text{Var}(X)$, where C is constant
- (D) If X and Y are independent random variables, then $E(XY) = E(X).E(Y)$
- (E) $\text{Var}(aX) = a.\text{Var}(X)$, where a is constant

Choose the correct answer from the options given below:

- (1) A, B and D only
- (2) B, D and E only
- (3) B, C and D only
- (4) B, C and E only

Q90. Which of the following are correct are OLS estimation in the presence of autocorrelation?

[December 2022 S1 · Q.48]

- (A) The estimated coefficients are linear
- (B) The estimated coefficients are unbiased
- (C) The estimated coefficients are biased
- (D) The estimated coefficients are consistent
- (E) The estimated coefficients are efficient

Choose the correct answer from the options given below:

- (1) A, C and D only
- (2) A, B and D only
- (3) A, C and E only
- (4) C, D and E only

Q91. Which of the following are correct in case of area under the standard normal probability curve. [December 2022 S1 · Q.49]

- (A) Distance from the mean ordinates in terms of $\pm \sigma$ at $Z = \pm 0.6745$ is 0.55
- (B) Distance from the mean ordinates in terms of $\pm \sigma$ at $Z = \pm 1.96$ is 0.95
- (C) Distance from the mean ordinates in terms of $\pm \sigma$ at $Z = \pm 2.0$ is 0.9544
- (D) Distance from the mean ordinates in terms of $\pm \sigma$ at $Z = \pm 2.58$ is 0.98
- (E) Distance from the mean ordinates in terms of $\pm \sigma$ at $Z = \pm 3.0$ is 0.9973

Choose the correct answer from the options given below:

- (1) A, C and E only
- (2) B, C and E only
- (3) B, C and D only
- (4) A, B and E only

Q92. Which of the following satisfy the assumptions of the classical linear regression model?

[December 2022 S1 · Q.50]

- (A) The regression model is linear in the parameters
- (B) Zero mean value of disturbance U_i
- (C) The variance of U_i is not constant
- (D) There is a serial autocorrelation between the disturbance terms
- (E) The number of observations must be greater than the number of parameters to be estimated

Choose the correct answer from the options given below:

- (1) B, C and E only
- (2) A, B and E only
- (3) B, D and E only
- (4) A, C and D only

Q93. If A, B and C are three different non-empty sets, then find which of the following are correct in laws of the set operation. [December 2022 S1 · Q.51]

- (A) $A \cap \emptyset = \emptyset$
- (B) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
- (C) $A \cap A' = \emptyset$
- (D) $B - A = B' \cap A'$
- (E) $(A \cup B)' = A' \cup B'$

Choose the correct answer from the options given below:

- (1) A, B, C and E only
- (2) A, C and E only
- (3) A, C and D only
- (4) A, B and C only

Q94. Which of the following are correct in case of probability of two independent events A and B?

[December 2022 S1 · Q.59]

- (A) $P(A/B) = P(A)$
- (B) $P(A/B) = P(B)$
- (C) $P(A \cap B) = P(A) \cdot P(B)$
- (D) $P(\bar{A} \cap B) = P(\bar{A}) \cdot P(B)$
- (E) $P(A \cap B) = P(\bar{A}) \cdot P(B)$

Choose the correct answer from the options given below:

- (1) B, C and E only
- (2) A, C and D only
- (3) B, C and D only
- (4) A, C and E only

Q95. Match List I with List II [December 2022 S1 · Q.68]

List - I	List - II
(A) Censored regression Model	(I) Probit Model
(B) Normit Model	(II) Marc Nerlove
(C) Distributed-lags Model	(III) Tobit Model
(D) Partial adjustment Model	(IV) L. M. Koyck

Choose the correct answer from the options given below:

- (1) A - I, B - III, C - II, D - IV
- (2) A - III, B - I, C - IV, D - II
- (3) A - III, B - IV, C - I, D - II
- (4) A - IV, B - III, C - I, D - II

Q96. Match List I with List II [December 2022 S1 · Q.69]

List - I	List - II
(A) $Y = \beta_1 + \beta_2 X$	(I) $\beta_2 X$
(B) $\ln Y = \beta_1 + \beta_2 X$	(II) β_2
(C) $Y = \beta_1 + \beta_2 \ln X$	(III) $\beta_2 (X/Y)$
(D) $\ln Y = \beta_1 + \beta_2 \ln X$	(IV) $\beta_2 (1/Y)$

(List I: Two-Variable Linear Regression Model; List II: Elasticity)

Choose the correct answer from the options given below:

- (1) A - III, B - IV, C - I, D - II
- (2) A - III, B - I, C - IV, D - II
- (3) A - II, B - I, C - III, D - IV
- (4) A - II, B - III, C - IV, D - I

UGC NET December 2022 (Shift 2)

Q97. If the two events A and B are not independent then which of the following is correct? [December 2022 S2 · Q.10]

- (1) $P(A \cap B) = P(A) \cdot P(B)$
- (2) $P(A \cap B) = P(A) \cdot P(B/A)$
- (3) $P(A \cap B) = P(A) \cdot P(A/B)$
- (4) $P(A \cap B) = 1 - P(A \cup B)$

Q98. In logit model, as the odds ratio decreases from 1 to 0, the logit becomes [December 2022 S2 · Q.16]

- (1) Negative
- (2) Equal to 0
- (3) Positive
- (4) Fraction

Q99. Given regression model as $Y_i = \alpha_1 + \beta_1 X_i + \beta_2 (X_i - X^*) D_i + U_i$ such a model is used to: [December 2022 S2 · Q.17]

- (1) Deseasonalise a quarterly time series data
- (2) Seasonally adjust a monthly time series data
- (3) To conduct chow test for structural break in time series data
- (4) Test piecewise linear relationship in the data

Q100. A player tosses two fair coins. He wins Rs. 5 if 2 heads occur, Rs 2 if 1 head occurs and Rs. 1 if no head occurs. Find his expected gain. [December 2022 S2 · Q.18]

- (1) Rs. 2
- (2) Rs. 2.25
- (3) Rs. 2.75
- (4) Rs. 2.50

Q101. A card is thrown from a well shuffled pack of playing cards. Find the probability that it is either a diamond or a king. [December 2022 S2 · Q.19]

- (1) 17/52
- (2) 16/52
- (3) 18/52
- (4) 15/52

Q102. In simultaneous equation models, OLS can be applied directly if [December 2022 S2 · Q.20]

- (1) It is a recursive model
- (2) Order condition is satisfied
- (3) Rank condition is satisfied
- (4) Both order and rank conditions are satisfied

Q103. Which of the followings are correct in case of Poisson distribution ? [December 2022 S2 · Q.47]

- (A) 'n' the number of trials is indefinitely large i.e. $n \rightarrow \infty$
- (B) 'p', the constant probability of success for each trial is indefinitely large
- (C) $np = m$ is finite
- (D) $\mu_3 = m+2$
- (E) $\mu_4 = m + 3m^2$

Choose the correct answer from the options given below:

- (1) A, C and D only
- (2) A, C and E only
- (3) B, C and E only
- (4) B, C and D only

Q104. Which of the followings are correct conclusions without error in case of testing of Hypothesis ? [December 2022 S2 · Q.48]

- (A) Reject null hypothesis (H_0), when it is true
- (B) Accept H_0 , when it is true
- (C) Accept H_0 , when Alternate hypothesis (H_1) is true.
- (D) Reject H_0 when H_1 is true.
- (E) Accept H_0 when H_1 is false.

Choose the correct answer from the options given below:

- (1) A, B and D only
- (2) B, D and E only
- (3) B, C and E only
- (4) A, C and E only

Q105. Which of the followings are correct for OLS estimation in the presence of Heteroscedasticity ? [December 2022 S2 · Q.49]

- (A) The estimated coefficients are linear
- (B) The estimated coefficients are unbiased
- (C) The estimated coefficients are biased
- (D) The estimated coefficients are consistent
- (E) The estimated coefficients are efficient.

Choose the correct answer from the options given below:

- (1) A, C and D only
- (2) A, B and D only
- (3) A, C and E only
- (4) A, C and D only

Q106. Which of the followings satisfy the assumptions of the classical linear regression model ?

[December 2022 S2 · Q.50]

- (A) Zero mean value of disturbance U_i
- (B) The explanatory variables and the error terms are not independent
- (C) Homoscedasticity or constant variance of U_i
- (D) No auto correlation between the disturbances
- (E) There is more than one exact linear relationship between explanatory variables.

Choose the most appropriate answer from the options given below:

- (1) A, B, C and D only
- (2) A, C, D and E only
- (3) A, C and D only
- (4) C, D and E only

Q107. If A, B and C are three different non-empty sets, then find which of the following are correct in laws of set operations ? [December 2022 S2 · Q.51]

- (A) $(A')' = A$
- (B) $(A \cup B) = 1 - (A \cap B)$
- (C) $A \cup (B \cap C) = (A \cup B) \cap C$
- (D) $A \cup (B \cap C) = (A \cap B) \cup (A \cap C)$
- (E) $A - B = A \cap B'$

Choose the correct answer from the options given below:

- (1) A, C and D only
- (2) A, C and E only
- (3) A, B and C only
- (4) C, D and E only

Q108. In multiple regression model, a hypothesis such as $H_0 : \beta_2 = \beta_3 = 0$ can be tested using

[December 2022 S2 · Q.52]

- (1) t-test
- (2) Chi-square test
- (3) ANOVA test
- (4) F-test

Q109. Match List I with List II [December 2022 S2 · Q.68]

List - I	List - II
(A) Detecting Serial Correlation	(I) Goldfeld - Quandt test
(B) Detection of Heteroscedasticity	(II) The Run's test
(C) Detection of Multicollinearity	(III) Ramsay's RESET test
(D) Test of specification error	(IV) Condition Index

Choose the correct answer from the options given below:

- (1) A-III, B-I, C-IV, D-II
- (2) A-II, B-IV, C-I, D-III
- (3) A-II, B-I, C-IV, D-III
- (4) A-IV, B-I, C-III, D-II

Q110. Match List I with List II (List II items are graphs with y on the vertical axis and x on the horizontal axis, origin 0) [December 2022 S2 · Q.69]

List - I	List - II
(A) Linear function	(I) [graph: curve starting at a positive y-intercept and rising at an increasing rate (convex upward-sloping curve)]
(B) Exponential function	(II) [graph: inverted-U curve starting from the origin, rising to a peak and then falling]
(C) Quadratic function	(III) [graph: wave-shaped curve that rises to a peak, falls to a trough and then rises again]
(D) Cubic function	(IV) [graph: straight upward-sloping line with a positive y-intercept]

Choose the correct answer from the options given below:

- (1) A-IV, B-II, C-I, D-III
- (2) A-I, B-IV, C-III, D-II
- (3) A-IV, B-I, C-II, D-III
- (4) A-IV, B-III, C-II, D-I

UGC NET June 2023 (20 June 2023)

Q111. The merchant's file of 20 accounts contains 6 delinquent and 14 non-delinquent accounts. An auditor randomly selects 5 of these accounts for examination. What is the probability that the auditor finds exactly 2 delinquent cases? [June 2023 · Q.60]

- (1) 0.2562
- (2) 0.3
- (3) 0.3087
- (4) 0.4526

Q112. A company has 140 employees, of which 30 are supervisors. 80 of the employees are married, and 20% of the married employees are supervisors. If a company employee is randomly selected, what is the probability that the employee is married and is a supervisor? [June 2023 · Q.61]

- (1) 0.1531
- (2) 0.1253
- (3) 0.0923
- (4) 0.1143

Q113. The Hausman's specification error test is used to test whether [June 2023 · Q.62]

- (1) An exogenous variable is correlated with the error term.
- (2) An endogenous variable is correlated with the error term.
- (3) Both an exogenous variable is correlated with the error term and an endogenous variable is correlated with the error term.
- (4) OLS method is appropriate to estimate the SEM.

Q114. In random walk without drift, [June 2023 · Q.63]

- (1) The effect of shock persists throughout the time period.
- (2) The effect of shock in the past dies out over time.
- (3) The effect of shock drifts away quickly.
- (4) There is no effect of past shock.

Q115. Given below are the alternative formulae for F-test statistics. We are interested in testing the statistical significance of the incremental contribution of X_3 at 5% level of significance to $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + U$. Which of these is not appropriate? Where RSS is residual sum of square and ESS is explained sum of square. [June 2023 · Q.64]

- (1) $F = [R^2_{UR} - R^2_R / d.f.] / [(1 - R^2_{UR}) / d.f.]$
- (2) $F = [ESS / d.f.] / [RSS / d.f.]$
- (3) $F = [ESS_{new} - ESS_{old} / d.f.] / [ESS_{new} / d.f.]$
- (4) $F = [R^2 / d.f.] / [(1 - R^2) / d.f.]$

Q116. Integrate : $\int x\sqrt{x^2 + 1} dx, x > 0$ [June 2023 · Q.65]

- (1) $(1/2)(x^2 + 1)^{3/2} + C$
- (2) $(1/3)(x^2 + 1)^{3/2} + C$
- (3) $(1/3)(x^3 + 1)^{2/3} + C$
- (4) $(1/2)(x^3 + 1)^{2/3} + C$

Q117. Find the extremum of $u = x^2 + y^2 + z^2$ subject to $x + y + z = 1$. [June 2023 · Q.66]

- (1) $(\bar{x}, \bar{y}, \bar{z}) = (1/3, 1/2, 1/2)$
- (2) $(\bar{x}, \bar{y}, \bar{z}) = (1/2, 1/2, 1/3)$
- (3) $(\bar{x}, \bar{y}, \bar{z}) = (1/2, 1/2, 1/2)$
- (4) $(\bar{x}, \bar{y}, \bar{z}) = (1/3, 1/3, 1/3)$

Q118. Which of the following are correct in case of theory of sets. [June 2023 · Q.105]

- (A) $(A \cup B') \cap (A' \cup C) \cap (B \cup C') = (A \cap B \cap C) \cup (A \cup B \cup C)$
- (B) $(A \cup B') \cap (A' \cup C) \cap (B \cup C') = (A \cap B \cap C) \cup (A \cup B \cup C)'$
- (C) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
- (D) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
- (E) $(A \cup B') \cap (A' \cup C) \cap (B \cup C') = (A \cap B \cap C) \cup (A' \cap B' \cap C')$

Choose the correct answer from the option given below :

- (1) A, C, E only
- (2) B, D, E only
- (3) B, C, E only
- (4) A, B, D only

Q119. If S is a sample space and E is an event then the probability axioms can be written as [June 2023 · Q.106]

- (A) $P(A) \geq 0$
- (B) $P(S) = 1$
- (C) $P(\cup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} P(A_i)$
- (D) $P(A) \leq 0$
- (E) $P(\cup_{i=1}^{\infty} S_i) = \sum_{i=1}^{\infty} P(S_i)$

Choose the correct answer from the option given below :

- (1) A and B only
- (2) A, B and D only
- (3) A, B and E only
- (4) A, B and C only

Q120. Properties of sufficient estimators are [June 2023 · Q.107]

- (A) It may not be unbiased
- (B) It may be unbiased
- (C) It is always consistent
- (D) It is not always consistent
- (E) Minimum variance unbiased estimator

Choose the correct answer from the option given below :

- (1) A, B and C only
- (2) A, C and E only
- (3) A, B, C and D only
- (4) A, B, C and E only

Q121. Which of the following are not symptoms of multicollinearity in a regression model. [June 2023 · Q.110]

- (A) High R^2 with few significant t ratio for coefficients.
- (B) High pair-wise correlations among regressors.
- (C) The closer the Tolerance (TOLj) is to zero.
- (D) Variance Inflation Factor (VIF) of a variable is below 10.
- (E) The closer the Tolerance (TOLj) is to 1.

Choose the correct answer from the option given below :

- (1) C and D only
- (2) D and E only
- (3) A and C only
- (4) B and D only

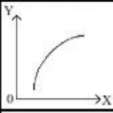
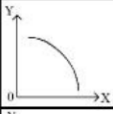
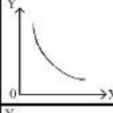
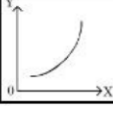
Q122. Match List I with List II [June 2023 · Q.117]

List - I	List - II
(A) Kleins rule of thumb	(I) Structural break
(B) Ljung Box (LB) statistic	(II) Unit Root
(C) KPSS	(III) Significance of autocorrelation coefficients
(D) Bai-Perron	(IV) Multicollinearity

Choose the correct answer from the options given below:

- (1) A-IV, B-III, C-II, D-I
- (2) A-IV, B-III, C-I, D-II
- (3) A-II, B-IV, C-I, D-III
- (4) A-II, B-III, C-IV, D-I

Q123. Match List I with List II [June 2023 · Q.124]

LIST I		LIST II	
A.	$f'(x) > 0$ and $f''(x) > 0$	I.	
B.	$f'(x) > 0$ and $f''(x) < 0$	II.	
C.	$f'(x) < 0$ and $f''(x) < 0$	III.	
D.	$f'(x) < 0$ and $f''(x) > 0$	IV.	

Choose the correct answer from the options given below:

- (1) A-I, B-IV, C-III, D-II
- (2) A-I, B-III, C-II, D-IV
- (3) A-IV, B-I, C-II, D-III
- (4) A-IV, B-II, C-III, D-I

Q124. Sequence of steps followed in the estimation of Indirect Least Square (ILS) starting from the beginning : [June 2023 · Q.133]

- (A) Identification of the structural equations
- (B) Application of OLS
- (C) Obtaining structural coefficients
- (D) Obtaining reduced form equation
- (E) Use the model for policy

Choose the correct answer from the option given below :

- (1) A, D, B, C and E
- (2) A, B, D, C and E
- (3) B, D, A, E and C
- (4) B, A, C, D and E

UGC NET December 2023 (12 Dec 2023, Shift 1)

Q125. In a regression of a dependent variable on intercept and k other independent variables using n observations, what are respectively the numerator and denominator degrees of freedom for the F test for overall significance of the regression ? [December 2023 S1 · Q.53]

- (1) $k - 1, n - k$
- (2) $k, n - k - 1$
- (3) $k - 1, n - k - 1$
- (4) $k, n - k$

Q126. Suppose X_i is income of an individual i and Y_i is the event that this individual owns a house, $P_i = \text{prob}(Y_i = 1 | X_i)$ and $Z_i = \beta_1 + \beta_2 X_i$, which of the following is true for the logit model ?

[December 2023 S1 · Q.54]

- (1) $P_i = 1 / (1 - e^{-Z_i})$
- (2) $P_i = 1 / (1 + e^{Z_i})$
- (3) $P_i = 1 / (1 + e^{-Z_i})$
- (4) $P_i = 1 / (1 - e^{Z_i})$

Q127. If the estimated slope coefficient in the regression of Y on X is β , what will be the estimated slope coefficient in the regression of 5 Y on 10 X ? [December 2023 S1 · Q.55]

- (1) 5β
- (2) 10β
- (3) 2β
- (4) 0.5β

Q128. Which of the following tests has stationarity as the null hypothesis ? [December 2023 S1 · Q.56]

- (1) ADF Test
- (2) Phillips Perron Test
- (3) ERS Test
- (4) KPSS Test

Q129. For a VAR model estimated using two variables X and Y, Lags of Y are jointly significant in the equation for X but Lags of X are jointly insignificant in the equation for Y. What can you conclude from this ? [December 2023 S1 · Q.57]

- (1) There is no Granger causality between X and Y
- (2) There is bidirectional Granger causality between X and Y
- (3) X Granger causes Y but Y does not Granger cause X
- (4) Y Granger causes X but X does not Granger cause Y

Q130. In the model $\ln Y_i = \alpha + \beta X_i + u_i$ Where $\ln$ is natural log, the elasticity of Y w.r.t. X is given by : [December 2023 S1 · Q.59]

- (1) β
- (2) βX_i
- (3) β / Y_i
- (4) $\beta X_i / Y_i$

Q131. Suppose you are testing a Null Hypothesis about the population mean, $H_0 : \mu = \mu_0$, against $H_1 : \mu > \mu_0$, with population variance given. The computed value of the test statistic is Z_c . What is the p-value for this test, where $\Phi(\cdot)$ is the cumulative distribution function for the standard normal distribution ? [December 2023 S1 · Q.65]

- (1) $\Phi(Z_c)$
- (2) $\Phi(Z_c/2)$
- (3) $1 - \Phi(Z_c)$
- (4) $1 - \Phi(Z_c/2)$

Q132. The power of a test refers to : [December 2023 S1 · Q.66]

- (1) Type I error
- (2) Type II error
- (3) $1 - \text{Type II error}$
- (4) $1 - \text{Type I error}$

Q133. Given a random sample of size n from a population with mean μ and variance σ^2 , the sample mean has the following expected value and variance : [December 2023 S1 · Q.67]

- (1) $E(\bar{x}) = \mu, \text{Var}(\bar{x}) = \sigma^2/n$
- (2) $E(\bar{x}) = n\mu, \text{Var}(\bar{x}) = n\sigma^2$
- (3) $E(\bar{x}) = \mu, \text{Var}(\bar{x}) = \sigma^2$
- (4) $E(\bar{x}) = \mu, \text{Var}(\bar{x}) = n/\sigma^2$

Q134. In March, 2023 the consumer price index (CPI) of a country with a 1982 base index of 100, was 234.7. This means the cost of purchasing the market basket of goods and services : [December 2023 S1 · Q.68]

- (1) has decreased by 134.7% since 1982
- (2) has increased by 34.7% since 1982
- (3) has increased by 234.7% since 1982
- (4) has increased by 134.7% since 1982

Q135. If $f(x, y)$ is a homogeneous function of degree 2 with $f'(2, 3) = 4$ and $f'(4, 6) = 12$, find $f(6, 9)$: [December 2023 S1 · Q.71]

- (1) 117
- (2) 234
- (3) 115
- (4) 243

Q136. Suppose $f(\cdot)$ is a continuous function over a closed and bounded interval $[a, b]$. Then there exists a point d in $[a, b]$ where $f(\cdot)$ has a minimum, and a point c in $[a, b]$ where $f(\cdot)$ has a maximum, so that $f(d) \leq f(x) \leq f(c)$ for all x in $[a, b]$. The statement is derived from : [December 2023 S1 · Q.74]

- (1) Intermediate Value Theorem
- (2) Rolle's Theorem
- (3) Mean value Theorem
- (4) Extreme value Theorem

Q137. If $A = \begin{bmatrix} 2 & 3 \\ 0 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 0 \\ 1 & 5 \end{bmatrix}$ which of the following is not correct ? [December 2023 S1 · Q.81]

- (1) $A^T B = \begin{bmatrix} 4 & 0 \\ 10 & 20 \end{bmatrix}$
- (2) $(A + B)^T = \begin{bmatrix} 4 & 1 \\ 3 & 9 \end{bmatrix}$
- (3) $A^T B = \begin{bmatrix} 0 & 4 \\ 10 & 20 \end{bmatrix}$
- (4) $A^T B^T = \begin{bmatrix} 4 & 2 \\ 6 & 23 \end{bmatrix}$

Q138. Which of the following statements are not true for the Linear Probability Model (LPM) ? [December 2023 S1 · Q.92]

- (A) Conditional probability of occurrence of an event conditional on the value of explanatory variable X is a linear function of X
- (B) Estimated value of the conditional probability of occurrence of the event lies between 0 and 1
- (C) Error term is homoscedastic
- (D) R^2 is a good measure of goodness of fit

Choose the correct answer from the options given below :

- (1) (B) Only
- (2) (B) and (C) Only
- (3) (B), (C) and (D) Only
- (4) (C) Only

Q139. Given that Z_t is a zero-mean white noise process with variance σ^2 , which of the following are $I(0)$ processes ? [December 2023 S1 · Q.93]

- (A) $X_t = X_{t-1} + Z_t$
- (B) $X_t = \alpha X_{t-1} + Z_t$ where $0 < \alpha < 1$
- (C) $X_t = Z_t + Z_{t-1}$
- (D) $X_t = 8 + X_{t-1} + Z_t$

Choose the correct answer from the options given below :

- (1) (C) Only
- (2) (B) Only
- (3) (B) and (C) Only
- (4) (B), (C) and (D) Only

Q140. Which of the following are features of a weakly stationary stochastic process ? [December 2023 S1 · Q.94]

- (A) Mean is time-varying
- (B) Mean is constant
- (C) Variance is constant
- (D) Covariance is time-invariant

Choose the correct answer from the options given below :

- (1) (B) and (C) Only
- (2) (B), (C) and (D) Only
- (3) (A), (C) and (D) Only
- (4) (A) and (C) Only

Q141. $A = \begin{bmatrix} 1 & 3 \\ 2 & 8 \\ 4 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 5 & 9 \end{bmatrix}$ [December 2023 S1 · Q.106]

- (A) AB is given by $\begin{bmatrix} 32 & 82 & 20 \end{bmatrix}$ (a column vector)
- (B) BA is given by $\begin{bmatrix} 32 & 82 & 20 \end{bmatrix}$
- (C) AB is not defined
- (D) BA is not defined
- (E) $A^T B$ is defined

Choose the correct answer from the options given below :

- (1) (C) and (B) Only
- (2) (A), (D), (E) Only
- (3) (A) Only
- (4) (A) and (D) Only

Q142. The OLS regression output of wages (Rs in thousand) on education (years of schooling S) and cognitive ability scores (CA) comes out to be : [December 2023 S1 · Q.108]

$$\text{wages}^{\wedge} = -3.36 + 0.64 S + 0.15 CA$$

(1.98) (0.12) (0.05)

with figures in parenthesis giving standard error.

The explained sum of squares is 4745.75 with degrees of freedom 2 and residual sum of squares is 33651.28 with degrees of freedom 567, which of the following statements is/are true ?

(A) For one year increase in schooling, wage per hour increases by Rs. 640 on the average, when CA is kept constant.

(B) The value of t statistic of CA is 3.

(C) The F statistic is given by $F = (4745.75/2) / (33651.28/567)$

(D) The F statistic is given by $F = (33651.28/567) / (4745.75/2)$

Choose the correct answer from the options given below :

(1) (A), (C) Only

(2) (A), (B), (D) Only

(3) (A), (B), (C) Only

(4) (A), (D) Only

Q143. Consider a matrix $A = \begin{bmatrix} 3 & 5 \\ -2 & -4 \end{bmatrix}$ [December 2023 S1 · Q.110]

consider the following statements on characteristic polynomial, Eigen Values and Eigen Vectors.

(A) The characteristic polynomial is $\lambda^2 + \lambda - 2$

(B) The characteristic polynomial is $\lambda^2 - 2\lambda - 4$ and the characteristic root or Eigen Values are $\lambda^* = 2$ and 1

(C) The characteristic polynomial is $\lambda^2 - \lambda + 2$ and the characteristic root or Eigen Values are $\lambda^* = 1$ and 2

(D) The characteristic root or Eigen Values are $\lambda^* = -2$ and 1

Choose the correct answer from the options given below :

(1) (A), (C) and (D) Only

(2) (B) and (D) Only

(3) (B) and (C) Only

(4) (A) and (D) Only

Q144. Consider the following statements : [December 2023 S1 · Q.112]

(A) The union of finite numbers of closed set is a closed set

(B) The empty set is an open set but not a closed set

(C) Every infinite set is an open set

(D) The intersection of finite numbers of open sets is an open set

Choose the correct answer from the options given below :

(1) (A) and (B) Only

(2) (B) and (C) Only

(3) (A) and (D) Only

(4) (A), (B) and (C) Only

Q145. Consider the following 3 Constrained Optimization Problems (COP). [December 2023 S1 · Q.115]

- (a) Maximize $z = \ln x$, subject to $x > 1$
- (b) Maximize $z = \ln x$, subject to $x < 1$
- (c) Maximize $z = \ln x$, subject to $1 < x < 2$
- (A) (a) and (b) have no solutions but (c) has solution
- (B) (a) and (b) have solutions but (c) has no solution
- (C) (a), (b) and (c) have no solution
- (D) (a) and (c) have no solution but (b) has solution

Choose the correct answer from the options given below :

- (1) (A) Only
- (2) (B) Only
- (3) (C) Only
- (4) (D) Only

Q146. Match List - I with List - II. [December 2023 S1 · Q.120]

List - I	List - II
List - I: Model/method	List - II: Feature
(A) Logit model	(I) elasticities are constant
(B) Log-log model	(II) binary dependent variable
(C) Autoregressive model	(III) corrects for heteroscedasticity
(D) Weighted least squares	(IV) includes lags of dependent variable

Choose the correct answer from the options given below :

- (1) A-II, B-I, C-III, D-IV
- (2) A-I, B-II, C-IV, D-III
- (3) A-I, B-II, C-III, D-IV
- (4) A-II, B-I, C-IV, D-III

Q147. Match List - I with List - II. [December 2023 S1 · Q.123]

List - I	List - II
List - I: Variable	List - II: Distribution
(A) Sum of squares of n independent standard normal variates	(I) t distribution
(B) Ratio of a standard normal variate to square root of an independent, chi square variable (divided by its degrees of freedom)	(II) F distribution
(C) Ratio of 2 independent chi-square variables each divided by their respective degrees of freedom	(III) Standard normal distribution
(D) Normal variate minus its mean, divided by its standard deviation	(IV) chi square distribution

Choose the correct answer from the options given below :

- (1) A-I, B-IV, C-II, D-III
- (2) A-IV, B-I, C-III, D-II
- (3) A-I, B-IV, C-III, D-II
- (4) A-IV, B-I, C-II, D-III

Q148. Chow Test is normally run to understand if a single regression run on pooled data is more efficient to describe a relation between a dependent variable and a set of independent variables than the same regression being run separately on two subgroups of sample. The steps involved in running the Chow Test are : [December 2023 S1 · Q.131]

- (A) Run the regression model on pooled sample and calculate RSS_p
- (B) Run the same regression model on the two sub-samples separately and calculate RSS_1 and RSS_2
- (C) Calculate the F statistic
- (D) Check the critical F with related degrees of freedom and compare it with F calculated
- (E) Do not reject H_0 if $F_{\text{calculated}} < F_{\text{critical}}$

Choose the correct answer from the options given below :

- (1) (A), (B), (C), (D), (E)
- (2) (A), (D), (C), (B), (E)
- (3) (C), (A), (B), (D), (E)
- (4) (C), (B), (A), (D), (E)

Q149. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R). [December 2023 S1 · Q.139]

Assertion (A) : Given a random sample, the distribution of the sample mean tends to a normal distribution, as the sample size becomes very large.

Reason (R) : Central limit theorem

In the light of the above statements, choose the most appropriate answer from the options given below :

- (1) Both (A) and (R) are correct and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- (3) (A) is correct but (R) is not correct
- (4) (A) is not correct but (R) is correct

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Q150. Given $y = (1 + x)/(1 - 2x)$ find the correct alternative: [August 2024 S1 · Q.65]

- (1) $dy/dx = 3/(1 - 2x)$
- (2) $dy/dx = 3/(1 - 2x)^2$
- (3) $dy/dx = 3/(1 - 2x)^3$
- (4) $dy/dx = 3/(1 - x)^2$

Q151. The quadratic form $ax^2 + by^2 + 2hxy$ is negative for all values of x and y (other than $x = y = 0$) if and only if: [August 2024 S1 · Q.77]

- (1) $a > 0$ and $|a h; h b| < 0$
- (2) $a < 0$ and $|a h; h b| < 0$
- (3) $a < 0$ and $|a h; h b| > 0$
- (4) $a < 0$ and $|a h; h b| \neq 0$

Q152. ANCOVA models include regressors that are: [August 2024 S1 · Q.79]

- (1) Only quantitative variables
- (2) Only qualitative variables
- (3) Only categorical variables
- (4) Both qualitative and quantitative variables

Q153. A and B are two events such that $P(\bar{A}) = 0.4$ and $P(A \cap B) = 0.2$, then $P(A \cap B)$ is: [August 2024 S1 · Q.80]

- (1) 0.4
- (2) 0.2
- (3) 0.6
- (4) 0.8

Q154. You have carried out a regression analysis but after thinking about the relationship between the two variables, you have decided that you must swap the explanatory and response variables. After refitting the regression model to the data, you expect that: [August 2024 S1 · Q.81]

- (1) The value of correlation will change
- (2) The sign of slope will change
- (3) The value of coefficient of determination, will change
- (4) The value of SSE (sum of square errors) will change

Q155. The lowest value of a set of observations is 4.5 and their range is 10.9. Arithmetic mean is found to be 19.9 and median found to be 15.6. Which of the following statements is correct? [August 2024 S1 · Q.82]

- (1) Arithmetic mean is wrong, but the median is correct
- (2) Arithmetic mean is right, but the median is wrong
- (3) Both arithmetic mean and median are wrong
- (4) Both arithmetic mean and median are right

Q156. Estimation of regression coefficients in the presence of high but not perfect multicollinearity may result in all of these except: [August 2024 S1 · Q.83]

- (1) High confidence interval for the estimates
- (2) Almost all the estimates are statistically significant
- (3) A high R^2
- (4) Estimates are all BLUE

Q157. Position of points O (0, 0) and P (2, -3) in the region of graph of inequation $2x - 3y < 5$ will be: [August 2024 S1 · Q.84]

- (1) O inside and P outside
- (2) O and P both outside
- (3) O and P both inside
- (4) O outside and P inside

Q158. Testing for cointegration is performed by: [August 2024 S1 · Q.85]

- (1) Chow test
- (2) Phillips-Peron test
- (3) Engel-Granger test
- (4) Error-correction mechanism

Q159. Heteroscedasticity may arise due to various reasons. Which one of the following is not a reason? [August 2024 S1 · Q.86]

- (1) Extremely low or high values of X and Y co-ordinates in the data set
- (2) Correlation of variables over time
- (3) Incorrect specification of the functional form of the model
- (4) Incorrect transformation of variables

Q160. To test $H_0 : \mu = \mu_0$, when both μ and σ are unknown, the relevant test statistic is: [August 2024 S1 · Q.87]

- (1) $t_{(n-1)} = (\bar{x} - \mu_0)/(s'/\sqrt{n})$
- (2) $t_{(n-1)} = (\bar{x} - \mu_0)/(s'/\sqrt{(n-1)})$
- (3) $t_n = (\bar{x} - \mu_0)/(s'/\sqrt{n})$
- (4) $t_{(n-1)} = (\bar{x} - \mu_0)/(s'/n)$

Q161. Arrange the following expressions in terms of increasing order of magnitude [given: $a > 0$, $b > 0$, $x > 0$] [August 2024 S1 · Q.103]

- (A) $y = e^a$
- (B) $y = e^{a + bx}$
- (C) $y = e^{a + b/x}$
- (D) $y = e^{a - b/x}$
- (E) $y = e^{a + 2bx}$

Choose the correct answer from the options given below:

- (1) (D), (A), (C), (B), (E)
- (2) (D), (A), (B), (C), (E)
- (3) (D), (B), (C), (A), (E)
- (4) (E), (B), (C), (A), (D)

Q162. A feasible solution in a linear programming problem (LPP): [August 2024 S1 · Q.116]

- (A) Must satisfy all the problem's constraints simultaneously
- (B) Need not satisfy all the constraints, only some of them
- (C) Must be a corner point in the feasible region
- (D) may or may not optimise the value of the objective function
- (E) Must be greater than or equal to zero

Choose the correct answer from the options given below:

- (1) (A), (D), (E) only
- (2) (A), (D) only
- (3) (A), (E) only
- (4) (B), (C), (E) only

Q163. Match List - I with List - II. [August 2024 S1 · Q.131]

List - I	List - II
(A) Endogenous variable	(I) Significance of Autocorrelation coefficient
(B) Count R^2	(II) Simultaneous equation
(C) Ljung Box Statistic	(III) Panel data regression
(D) Least square dummy variable model	(IV) Logit regression

Choose the correct answer from the options given below:

- (1) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)
- (2) (A)-(II), (B)-(III), (C)-(I), (D)-(IV)
- (3) (A)-(II), (B)-(III), (C)-(IV), (D)-(I)
- (4) (A)-(II), (B)-(IV), (C)-(I), (D)-(III)

Q164. Match List - I with List - II. [August 2024 S1 · Q.136]

List - I	List - II
(A) General Heteroscedasticity test	(I) Ramsey
(B) General test for auto correlation	(II) Hausman
(C) Test of simultaneity	(III) White
(D) General test of specification error	(IV) Breusch-Godfrey

Choose the correct answer from the options given below:

- (1) (A)-(I), (B)-(II), (C)-(III), (D)-(IV)
- (2) (A)-(IV), (B)-(II), (C)-(I), (D)-(III)
- (3) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)
- (4) (A)-(III), (B)-(I), (C)-(II), (D)-(IV)

UGC NET December 2024**Q165. Match LIST-I with LIST-II** [December 2024 · Q.61]

List - I	List - II
(A) Type II error	(I) $P_{01} \times P_{10} = 1$
(B) Mean > Mode	(II) Not rejecting H_0 when H_0 is not true
(C) Homogenous population	(III) Positively skewed distribution
(D) Time reversal test	(IV) Simple random Sampling

Choose the correct answer from the options given below:

- (1) A-II, B-III, C-IV, D-I
- (2) A-III, B-I, C-II, D-IV
- (3) A-IV, B-II, C-III, D-I
- (4) A-I, B-II, C-III, D-IV

Q166. Match LIST-I with LIST-II [December 2024 · Q.81]

List - I	List - II
(A) Simple random sampling	(I) Non - probability Sampling
(B) Systematic sampling	(II) Random choice of items from each stratum
(C) Quota Sampling	(III) Random selection of the first unit and systematic selection of the rest
(D) Stratified random sampling	(IV) Equal probability of selection for each item in all trials

Choose the correct answer from the options given below:

- (1) A-IV, B-III, C-I, D-II
- (2) A-IV, B-II, C-I, D-III
- (3) A-III, B-II, C-I, D-IV
- (4) A-IV, B-III, C-II, D-I

Q167. The unit root tests for time series is based on the null hypothesis that the time series under consideration is - [December 2024 · Q.83]

- (1) Stationary
- (2) non - stationary
- (3) Strictly stationary
- (4) Weak stationary

Q168. Six number of fair dice are rolled simultaneously. What is the probability that each face shows up with a different number? [December 2024 · Q.86]

- (1) $1/6^6$
- (2) $6/6^6$
- (3) $6!/6^6$
- (4) $6^5/6^6$

Q169. Match LIST-I with LIST-II [December 2024 · Q.94]

List - I	List - II
(A) Durbin - Watson	(I) Multicollinearity
(B) Granger	(II) Heteroscedastic disturbances
(C) Farrar - Glauber	(III) Causality
(D) Glejser	(IV) Autocorrelated disturbances

Choose the correct answer from the options given below:

- (1) A-IV, B-III, C-I, D-II
- (2) A-IV, B-III, C-II, D-I
- (3) A-II, B-III, C-IV, D-I
- (4) A-III, B-IV, C-I, D-II

Q170. In a two variable regression, the dependent and independent variables are Y and X respectively. The coefficient of correlation between X and Y is 0.8. Which of the following statements is correct? [December 2024 · Q.105]

- (1) 8% of variation in Y is explained by X
- (2) 64% of variation in Y is explained by X
- (3) 0.8% of variation in Y is explained by X
- (4) 80% of variation in Y is explained by X

Q171. Consider the following system of equations : [December 2024 · Q.117]

$$Y_1 = \alpha_0 + \alpha_1 Y_2 + \alpha_3 Y_3 + \alpha_4 X_1 + \alpha_5 X_2 + U_1$$

$$Y_2 = \beta_0 + \beta_1 Y_3 + \beta_2 Y_1 + \beta_3 X_2 + U_2$$

$$Y_3 = \lambda_0 + \lambda_1 x_1 + \lambda_2 x_2 + \lambda_3 x_3 + U_3$$

According to the order condition, the first equation is :

- (1) Unidentified
- (2) Just identified
- (3) Over identified
- (4) Not possible to say because the reduced form of the model is not given

Q172. Given the objective function $y = f(x_1, x_2, \dots, x_n)$ and the constraint $g(x_1, x_2, \dots, x_n) = 0$, the second order condition for an extremum is : [December 2024 · Q.119]

- (A) $|H_2| < 0$, $|H_3| < 0$ for a maximum
- (B) $|H_2| > 0$, $|H_3| < 0$ for a maximum
- (C) $H_2 < 0$, $|H_3| > 0$ for a maximum
- (D) $|H_2| < 0$, $|H_3| < 0$ for a minimum
- (E) $|H_2| < 0$, $|H_3| < 0$ for a minimum

Choose the correct answer from the options given below:

- (1) A & D Only
- (2) C & E Only
- (3) B & D Only
- (4) B & E Only

Q173. Arrange the following events in increasing order of their probability of happening : [December 2024 · Q.120]

- (A) Probability of getting two heads in simultaneous toss of two coins
- (B) Probability of getting a queen from a single draw of a card from a well shuffled pack of cards
- (C) Probability of getting a white ball from a bag containing 3 white and 7 black balls
- (D) Probability of getting an even number from a single roll of a fair die

Choose the correct answer from the options given below:

- (1) A, D, B, C
- (2) C, A, B, D
- (3) B, C, A, D
- (4) B, A, C, D

Q174. Match LIST-I with LIST-II [December 2024 · Q.127]

List - I	List - II
(A) Unit Root test	(I) Durbin - Watson Test
(B) Contingency table	(II) Student's t-Test
(C) Regression coefficient	(III) Stationarity
(D) Autocorrelation	(IV) χ^2 Test (Chi square test)

Choose the correct answer from the options given below:

- (1) A-III, B-I, C-II, D-IV
- (2) A-III, B-IV, C-II, D-I
- (3) A-IV, B-II, C-III, D-I
- (4) A-I, B-III, C-II, D-IV

Q175. If a variable has 'm' categories, we can run the regression using : [December 2024 · Q.129]

- (1) 'm-1' dummy variables and an intercept term
- (2) 'm' dummy variables and an intercept term
- (3) '(m-1)' dummy variables without an intercept
- (4) '(m+1)' dummy variables without an intercept

Q176. The input - output matrix for a two sector economy is given by $A = \begin{bmatrix} 0.25 & 0.40 \\ 0.10 & 0.10 \end{bmatrix}$ (rows: [0.25 0.40] and [0.10 0.10]). If the external demand for the outputs of the two sectors is $D = \begin{bmatrix} 10 \\ 20 \end{bmatrix}$ (column vector), what will be the optional output levels of the two commodities? [December 2024 · Q.130]

- (1) 26.77 and 18.43
- (2) 9.33 and 24.50
- (3) 26.77 and 25.20
- (4) 24.33 and 25.20

Q177. Which of the following statements is true concerning the optimal solution of a linear programming problem with two decision variables? [December 2024 · Q.133]

- (1) There is always a unique solution to the problem.
- (2) The optimal solution is either an extreme point or is on the line connecting two extreme points.
- (3) All resources must be used up by an optimal solution.
- (4) The optimal solution may be an interior point of the set of feasible solutions.

Q178. Generalised least square method of estimation can be used in which of the following cases :

[December 2024 · Q.136]

- (A) Autocorrelated disturbances
- (B) Multicollinearity among explanatory variables
- (C) Heteroscedastic disturbances
- (D) Overfitted models
- (E) Incorrect functional forms

Choose the correct answer from the options given below:

- (1) A, C & D Only
- (2) C, B & E Only
- (3) A & C Only
- (4) C, D & E Only

Q179. Given $A = \begin{bmatrix} 2 & 0 & 4 \\ 1 & 3 & 1 \end{bmatrix}$ (a 2×3 matrix with rows (2 0 4) and (1 3 1)) and $B = (4 \ 5 \ 0)$, which of the following is true? [December 2024 · Q.139]

- (A) AB^T is (8 ; 19) (a column vector)
- (B) AB^T is defined
- (C) AB is not defined
- (D) BA is defined
- (E) BA^T is (8 19)

Choose the correct answer from the options given below:

- (1) A & E Only
- (2) A, B, C & E
- (3) A, B & C Only
- (4) A, C & E Only

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Q180. Bootstrapping technique is use to _____ [June 2025 S2 · Q.59]

- (1) Tests for specification bias
- (2) Obtain the sampling distribution of parameters of interest
- (3) Test for autocorrelation
- (4) Test for normality of error term

Q181. According to Granger and Newbold, a good rule of thumb to suspect that the estimated regression is spurious is given by: [June 2025 S2 · Q.60]

- (1) High R^2 with low t-value ($R^2 > t$)
- (2) High t values with low Durbin-Watson d value ($t > d$)
- (3) High R^2 and low Durbin-Watson d value ($R^2 > d$)
- (4) High R^2 and low F value ($R^2 > F$)

Q182. The indirect least square is applied to estimate the coefficient of the: [June 2025 S2 · Q.61]

- (1) Structural equation
- (2) Over identified equation
- (3) Reduced form equation
- (4) Under identified equation

Q183. The management of a manufacturing firm wishes to determine the average time required to complete a certain manual operation. There should be 0.95 confidence that error in the estimate will not exceed 2 minutes. What sample size is estimated by a time and motion study expert as 16 minutes? [June 2025 S2 · Q.62]

- (1) 96
- (2) 300
- (3) 255
- (4) 246

Q184. If x is a poisson variate with mean λ , then $P(x+1)$ is: [June 2025 S2 · Q.63]

- (1) $[\lambda/(x+1)] P(x)$
- (2) λ/x
- (3) $[(x+1)/\lambda] P(x)$
- (4) $(x/\lambda) P(x)$

Q185. A candidate is selected for interview for three posts. For the first post, there are 3 candidates, for the second 4 candidates and for the third 2 candidates. What is the probability that the candidate is selected for at least one post? [June 2025 S2 · Q.64]

- (1) $1/4$
- (2) $1/2$
- (3) $3/4$
- (4) $1/9$

Q186. The data about the sales and advertisement expenditure of a firm is given below: [June 2025 S2 · Q.65]

	Sales (In crore of Rs.)	Advertisement expenditure (In crore of Rs.)
Mean	40	6
Standard deviation	10	1.5

Coefficient of correlation = $r = 0.9$

Estimate the likely sales for the proposed advertisement of Rs. 10 crore of rupees.

- (1) Rs. 60 crore
- (2) Rs. 64 crore
- (3) Rs. 70 crore
- (4) Rs. 74 crore

Q187. If $\bar{x} = 32$, $\bar{y} = 38$, the regression coefficients $b_{xy} = -0.2337$, $b_{yx} = -0.6643$. Find the equation of the line of regression of y on x . [June 2025 S2 · Q.66]

- (1) $y = -0.2337x + 40.8806$
- (2) $y = -0.2337x - 40.8806$
- (3) $y = -0.6643x - 59.2576$
- (4) $y = -0.6643x + 59.2576$

Q188. If $U = 6x^2 + y^2$, and $4x - y = 1$. What is the value of U ? [June 2025 S2 · Q.67]

- (1) $4/15$
- (2) $3/11$
- (3) $3/16$
- (4) $4/21$

Q189. Integrate $\int (x+3)/(x+5) dx$. [June 2025 S2 · Q.68]

- (1) $x^2/2 - 2.\ln|x + 5| + c$
- (2) $x^2/2 + 2.\ln|x + 5| + c$
- (3) $x + 2.\ln|x + 5| + c$
- (4) $x - 2.\ln|x + 5| + c$

Q190. Evaluate $\int$ from 1 to 2 of $e^{-2x} dx$. [June 2025 S2 · Q.71]

- (1) $e^{-2} - e^{-4}$
- (2) $-(1/2)(e^{-2} - e^{-4})$
- (3) $(1/2)(e^{-2} - e^{-4})$
- (4) $(1/2)(e^{-2} + e^{-4})$

Q191. Assume that the rate of investment is described by the function $I(t) = 12t^{1/3}$ and capital stock $k(0) = 25$. Find the time path of capital stock k . [June 2025 S2 · Q.72]

- (1) $16t^{4/3} + 25$
- (2) $9t^{4/3} + 25$
- (3) $-9t^{4/3} + 25$
- (4) $-16t^{4/3} + 25$

Q192. The null hypothesis that all slope coefficients are simultaneously equal to zero is tested in logit model by. [June 2025 S2 · Q.77]

- (1) F-test
- (2) t-test
- (3) Chi-square test
- (4) Likelihood ratio statistics

Q193. Which of the followings are correct in case of Normal distribution? [June 2025 S2 · Q.111]

- (A) μ_3 for Normal distribution is 0.
- (B) β_2 for Normal distribution is 0.
- (C) Mean deviation (about mean) for Normal distribution is $(4/5)\sigma$
- (D) μ_4 for Normal distribution is $2\sigma^4$
- (E) Characteristic function for Normal distribution is $e^{i\mu t - t^2\sigma^2/2}$

Choose the correct answer from the options given below:

- (1) A, C & E Only
- (2) A, B, C & E Only
- (3) B, C & D Only
- (4) A, C & D Only

Q194. By using the uniqueness property of moment generating function (M.G.F.), which of the followings are correct distribution if M.G.F. is as follows? [June 2025 S2 · Q.112]

- (A) $M(t) = (1/2 + (1/2)e^t)^6$ is a poisson distribution.
- (B) $M(t) = (1 + e^t)^5 / 32$ is a binomial distribution.
- (C) $M(t) = e^{(3(e^t - 1))}$ is poisson distribution
- (D) $M(t) = e^{((e^t - 1)/4)}$ is poisson distribution
- (E) $M(t) = 4(3e^{-t} - 1)^{-2}$ is binomial distribution

Choose the correct answer from the options given below:

- (1) A, B, C & D Only
- (2) B, C & D Only
- (3) B, C, D & E Only
- (4) A, B, C, D & E

Q195. Match List - I with List - II [June 2025 S2 · Q.129]

List - I	List - II
(A) β_2 for Chi-square distribution	(I) $(n-2)$
(B) Skewness for Chi-square distribution	(II) $48n+12n^2$
(C) Mode for Chi-square distribution	(III) $(12/n)+3$
(D) μ_3 for Chi-square distribution	(IV) $\sqrt{(2/n)}$

Choose the correct answer from the options given below:

- (1) A-IV, B-III, C-II, D-I
- (2) A-III, B-II, C-IV, D-I
- (3) A-II, B-I, C-IV, D-III
- (4) A-III, B-IV, C-I, D-II

Q196. Match List - I with List - II [June 2025 S2 · Q.130]

List - I	List - II
(A) The mean of a Hypergeometric distribution	(I) q/p^2
(B) The moment generating function of negative binomial distribution	(II) $(Q - Pe^t)^{-r}$
(C) The coefficient of Kurtosis of a binomial distribution	(III) nM/N
(D) The variance of geometric distribution	(IV) $(1 - 6pq)/npq$

Choose the correct answer from the options given below:

- (1) A-II, B-III, C-I, D-IV
- (2) A-III, B-II, C-IV, D-I
- (3) A-I, B-III, C-IV, D-II
- (4) A-II, B-I, C-III, D-IV

Q197. Match List - I with List - II [June 2025 S2 · Q.131]

List - I	List - II
(A) $\int$ from 0 to ∞ of $1/(1+x^2)$ dx	(I) does not exist
(B) $\int$ from 1 to ∞ of $1/\sqrt{x}$ dx	(II) $1/3$
(C) $\int$ from $-\infty$ to -1 of $1/x^4$ dx	(III) $\pi/2$
(D) $\int$ from $-\infty$ to $+\infty$ of $1/(1+x^2)$ dx	(IV) π

Choose the correct answer from the options given below:

- (1) A-I, B-II, C-IV, D-III
- (2) A-IV, B-I, C-II, D-III
- (3) A-III, B-I, C-II, D-IV
- (4) A-II, B-III, C-IV, D-I

Passage / data for Q.141-145 (June 2025 S2)

Based on the information given below, answer the questions:

Two fair dice are thrown independently. Three events A, B, and C are defined as follows;

A : odd face with first dice

B : odd face with second dice

C : sum of points on two dice is odd

Q198. Find the $P(C)$. [June 2025 S2 · Q.141]

- (1) $1/4$
- (2) $1/2$
- (3) $3/4$
- (4) $5/6$

Q199. Find the $P(A \cap B)$. [June 2025 S2 · Q.142]

- (1) 0
- (2) $3/4$
- (3) $7/12$
- (4) $1/4$

Q200. Find the $P(A \cap C)$. [June 2025 S2 · Q.143]

- (1) 0
- (2) $1/2$
- (3) $1/6$
- (4) $1/4$

Q201. Find the $P(B \cap C)$. [June 2025 S2 · Q.144]

- (1) $1/2$
- (2) 0
- (3) $1/4$
- (4) $1/36$

Q202. Find the $P(A \cap B \cap C)$. [June 2025 S2 · Q.145]

- (1) 0
- (2) $1/2$
- (3) $1/4$
- (4) $1/6$

UGC NET December 2025 (Exam held 31 Dec 2025 - 7 Jan 2026)

Q203. Based on the following statements: [December 2025 · Q.56]

- (A) Matrix Addition is not Commutative
- (B) Matrix Addition is Associative
- (C) Matrix Multiplication is Commutative
- (D) Matrix Multiplication is Associative if product exists

Choose the correct answer from the options given below:

- (1) A and C only
- (2) A and D only
- (3) B and C only
- (4) B and D only

Q204. Given samples of sizes 20 and 15 with respective sample variances 35.53 and 34.92, the calculated value of F-statistic is: [December 2025 · Q.57]

- (1) 1.02
- (2) 0.98
- (3) 0.76
- (4) 1.31

Q205. Match List - I with List - II. [December 2025 · Q.58]

List - I	List - II
(A) Slack Variable	(I) Best of all feasible solutions
(B) Surplus Variable	(II) Solution violating at least one condition
(C) Optimal Solution	(III) Used when there are less than or equal to ($\leq$) type inequalities
(D) Infeasible Solution	(IV) Used when there are greater than or equal to ($\geq$) type inequalities

Choose the correct answer from the options given below:

- (1) A-IV, B-III, C-I, D-II
- (2) A-IV, B-III, C-II, D-I
- (3) A-III, B-IV, C-II, D-I
- (4) A-III, B-IV, C-I, D-II

Q206. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R). [December 2025 · Q.75]

Assertion (A): The presence of heteroscedasticity problem in regression analysis implies that the least square estimators are still unbiased but inefficient.

Reason (R): The estimates of the variances are also unbiased.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both (A) and (R) are correct and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- (3) (A) is correct but (R) is not correct
- (4) (A) is not correct but (R) is correct

Q207. Based on Beta (β) and Gama (γ) coefficients, if $\beta_2 < 3$ or $\gamma_2 < 0$ then the curve is said to be: [December 2025 · Q.76]

- (1) Normal Curve
- (2) Lepto Kurtic
- (3) Platy Kurtic
- (4) Insufficient Data

Q208. In a population of men in a city, 3% are observed to suffer from cancer, 15% are found to be smokers, while 10% are noticed to be either smokers or cancer patients. If a man is selected at random, what is the probability that he is a smoker and suffering from cancer? [December 2025 · Q.78]

- (1) 0.08
- (2) 0.8
- (3) 0.02
- (4) 0.2

Q209. Match List - I with List - II. [December 2025 · Q.79]

List - I	List - II
(A) Type I error	(I) Accept H_0 when it is false
(B) Type II error	(II) $H_0: \mu = \mu_0$ and $\sigma^2 = \sigma_0^2$
(C) Simple Hypothesis	(III) Reject H_0 when it is true
(D) Composite Hypothesis	(IV) $H_0: \mu = \mu_0$ and σ^2 is unknown

Choose the correct answer from the options given below:

- (1) A-III, B-I, C-IV, D-II
- (2) A-I, B-III, C-II, D-IV
- (3) A-III, B-I, C-II, D-IV
- (4) A-I, B-III, C-IV, D-II

Q210. If $P(A) = 1/5$ and $P(B) = 1/3$ and A and B are two mutually exclusive events, then the probability of neither A nor B will occur is: [December 2025 · Q.86]

- (1) $8/25$
- (2) $1/15$
- (3) $7/15$
- (4) $1/5$

Q211. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R). [December 2025 · Q.87]

If X is a given series of observations, a and b are any two constants, then with regard to Standard Deviation (SD).

Assertion (A): $SD(X + a) = SD(X - b) = SD(X)$ and $SD(aX) = a^2 \cdot SD(X)$

Reason (R): SD is independent of change of origin, but not of scale.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both (A) and (R) are correct and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- (3) (A) is correct but (R) is not correct
- (4) (A) is not correct but (R) is correct

Q212. Match List - I with List - II. [December 2025 · Q.99]

List - I	List - II
(A) nth term of an Arithmetic Progression	(I) $(n/2)[2a + (n - 1)d]$
(B) nth term of a Geometric Progression	(II) $a(r^n - 1)/(r - 1)$, $r > 1$
(C) Sum to n terms of an Arithmetic Progression	(III) $a + (n - 1)d$
(D) Sum to n terms of a Geometric Progression	(IV) $a r^{n-1}$

where a is first term, d is common difference and r is common ratio

Choose the correct answer from the options given below:

- (1) A-III, B-IV, C-II, D-I
- (2) A-III, B-IV, C-I, D-II
- (3) A-IV, B-III, C-I, D-II
- (4) A-IV, B-I, C-II, D-III

Q213. The range of the function $y = f(x) = \sqrt{1 - x^2}$ is: [December 2025 · Q.109]

- (1) [0, 1]
- (2) [1, 0]
- (3) [1, ∞]
- (4) [1, 2]

Q214. Given the difference equation $Y(t+1) - 1.2Y(t) = 0$, the general expression $Y(t)$ in terms of t can be: [December 2025 · Q.112]

- (1) $Y(t) = (1.2)^t Y_1$
- (2) $Y(t) = (1.2)^{t+1} Y_0$
- (3) $Y(t) = (1.2)^{t-1} Y_1$
- (4) $Y(t) = (1.2)^{t-1} Y_0 + C$

Q215. Given the structural model [December 2025 · Q.114]

$$y_1 = 3y_2 - 2x_1 + x_2 + u_1$$

$$y_2 = y_3 + x_3 + u_2$$

$$y_3 = y_1 - y_2 - 2x_3 + u_3$$

Based on the order condition, the second equation is:

- (1) over identified
- (2) exactly identified
- (3) under identified
- (4) information not sufficient

Q216. Coefficient of Determination (R^2): [December 2025 · Q.124]

- (A) is an increasing function of the number of regressors
- (B) is a decreasing function of the number of regressors
- (C) is unaffected by increasing or decreasing the number of regressors
- (D) is always positive

Choose the correct answer from the options given below:

- (1) A and B only
- (2) A and C only
- (3) A and D only
- (4) B and C only

Q217. Based on $\{\varepsilon_t\}$ is a purely random process with mean zero and variance σ^2 : [December 2025 · Q.131]

- (A) A process $\{x_t\}$ defined by $x_t = \alpha_1 x_{t-1} + \alpha_2 x_{t-2} + \dots + \alpha_r x_{t-r} + \varepsilon_t$ is called an autoregressive process of order r
- (B) A process $\{x_t\}$ defined by $x_t = \beta_0 \varepsilon_t + \beta_1 \varepsilon_{t-1} + \dots + \beta_m \varepsilon_{t-m}$ is called a moving average process of order m
- (C) A process $\{x_t\}$ defined by $x_t = \beta_0 \varepsilon_t + \beta_1 \varepsilon_{t-1} + \dots + \beta_m \varepsilon_{t-m}$ is called an autoregressive process of order r
- (D) A process $\{x_t\}$ defined by $x_t = \alpha_1 x_{t-1} + \alpha_2 x_{t-2} + \dots + \alpha_r x_{t-r} + \varepsilon_t$ is called a moving average process of order m

Choose the correct answer from the options given below:

- (1) A and B only
- (2) A and D only
- (3) B and D only
- (4) C and D only

Q218. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R). [December 2025 · Q.133]

Assertion (A): If X and Y are any two variables and are transformed to the new variables U and V defined by $U = (X - A)/h$, $V = (Y - B)/K$ where A, B, h, K are any constants, h and $K > 0$ then, $r_{xy} = r_{uv}$

Reason (R): Correlation coefficient(r) is independent of change of origin and scale.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both (A) and (R) are correct and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- (3) (A) is correct but (R) is not correct
- (4) (A) is not correct but (R) is correct